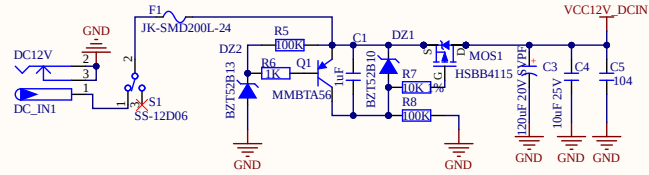


## Revision History

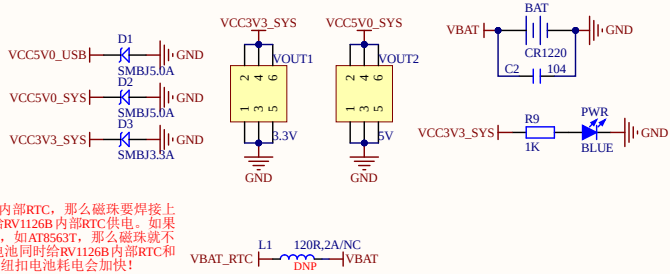
| Version | Date       | By          | Change Dscription         |
|---------|------------|-------------|---------------------------|
| V1.0    | 2025-12-31 | zuozhongkai | 1:Initial release version |
|         |            |             |                           |



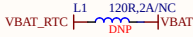
12V DCIN



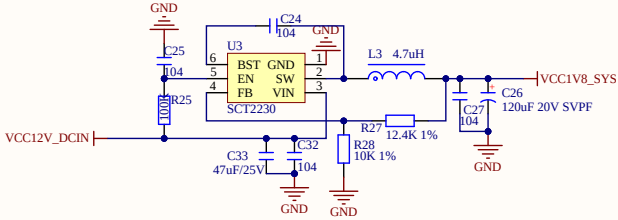
POWER IO



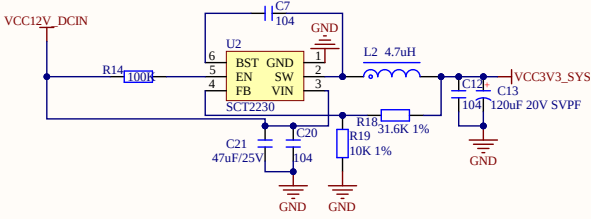
如果要使用RV1126B内部RTC，那么磁珠要焊接上，由底板纽扣电池给RV1126B内部RTC供电。如果要使用外部RTC芯片，如AT8563T，那么磁珠就不能焊接，否则纽扣电池同时给RV1126B内部RTC和外部RTC芯片供电，纽扣电池耗电会加快！



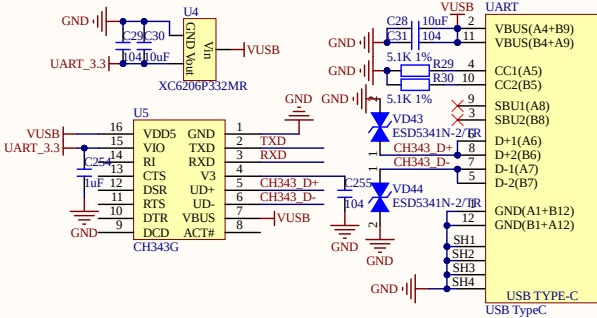
VCC1V8\_SYS



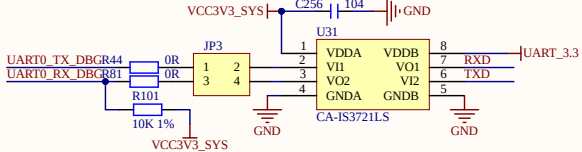
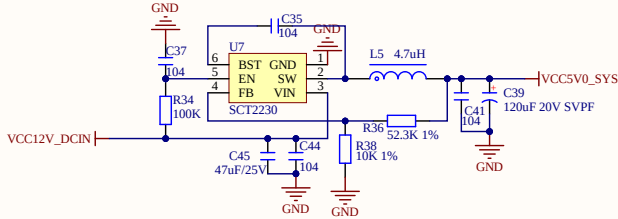
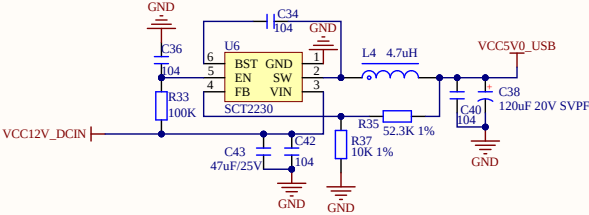
VCC3V3\_SYS



USB USART(TYPEC)

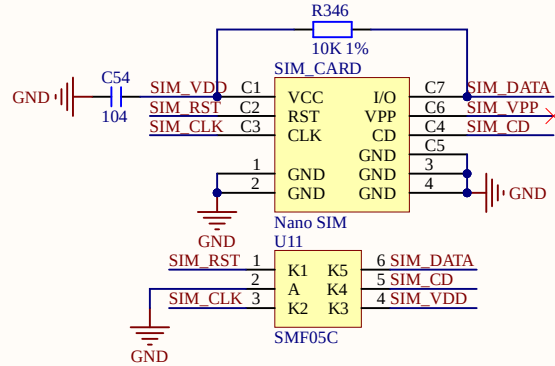
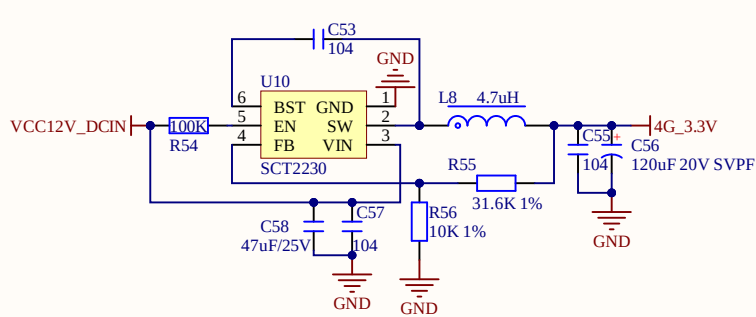
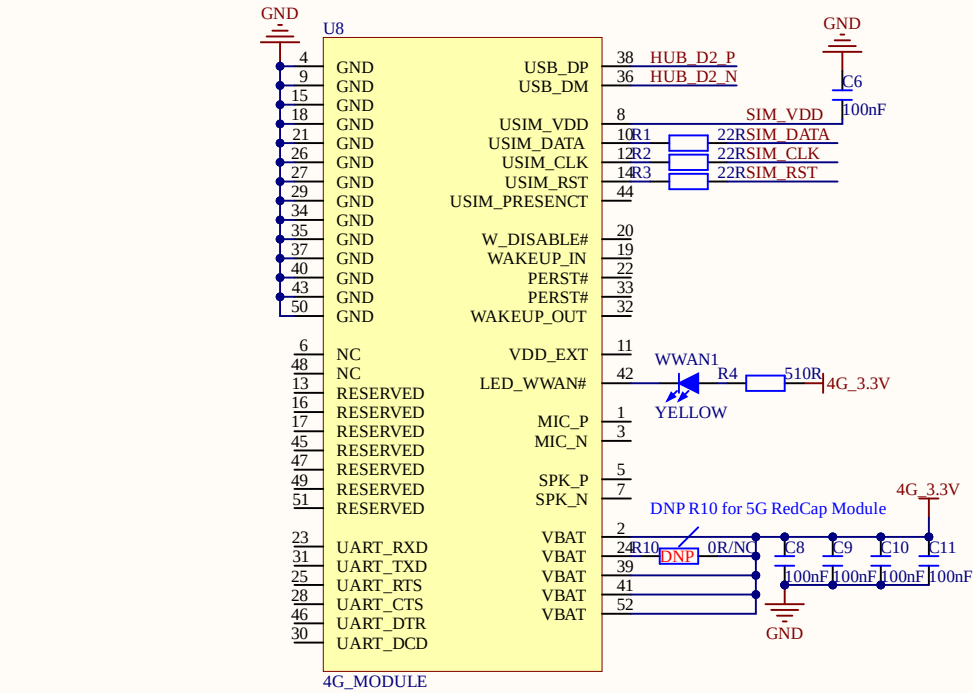


VCC5V0

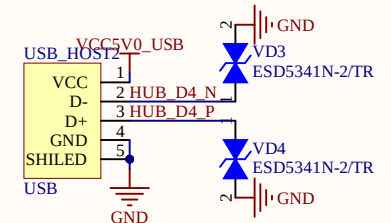
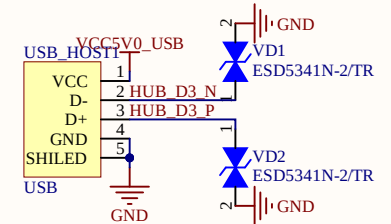
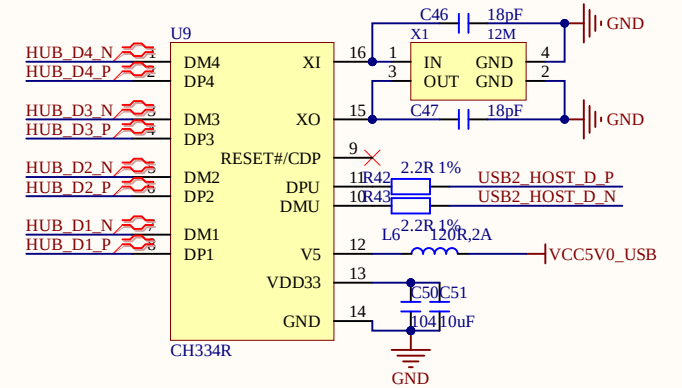


CA-IS3721LS是隔离保护芯片，CA-IS372x 器件具有高绝缘能力，有助于防止数据总线或其他电路上的噪声和浪涌进入本地接地端，从而干扰或损坏敏感电路。如果输入侧电源掉电或信号丢失，就会输出低电平。这里用来对开发板串口做保护，比如开发板正常工作时的时候拔掉串口USB，那么UART\_3.3就会掉电，此时CA-IS3721LS的B端输入就会进入保护状态，输出低电平，哪怕重新把串口USB插上也不会有串口数据传输。因此本开发板实际使用中不要在系统工作的时候插拔USB串口！！客户自己设计底板的时候可以根据实际使用场合，去掉这个隔离芯片

## 4G MODULE



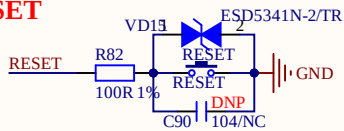
## USB HUB



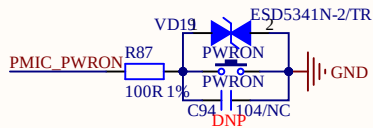
|                                        |                  |                |
|----------------------------------------|------------------|----------------|
| Title: 04_ATK_RV1126B_4G_USBHUB.SchDoc |                  |                |
| Project: ATK-DLVR1126B V1.0.PrjPcb     |                  |                |
| Size: A4                               | Author: ALIENTEK |                |
| Date: 2025-12-31                       | Version: V1.0    | Sheet: 4 of 13 |



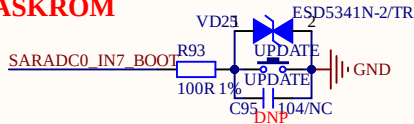
## RESET



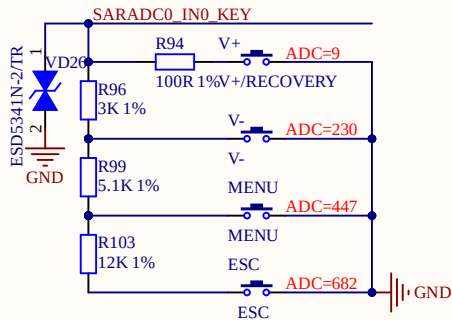
## PWRON



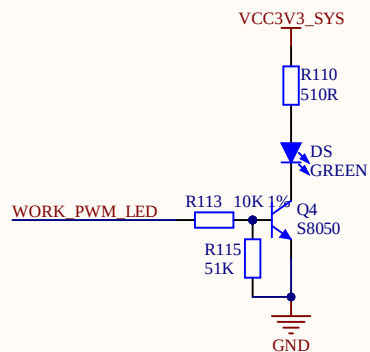
## MASKROM



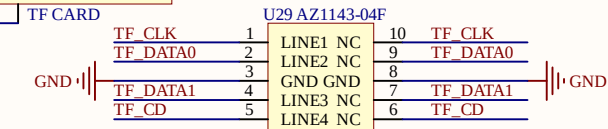
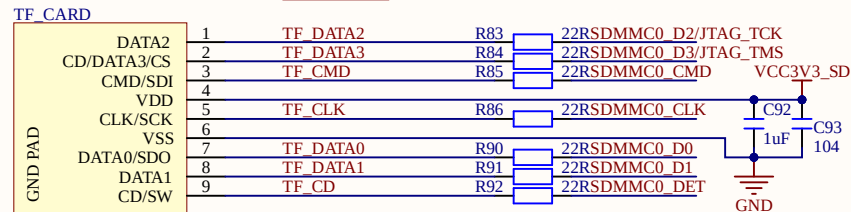
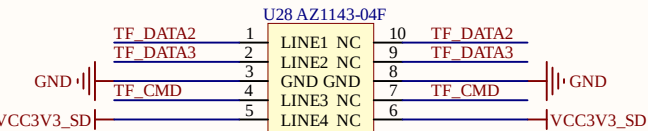
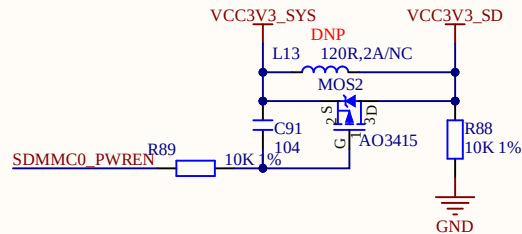
## KEY



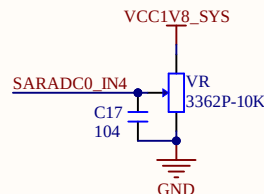
## LED



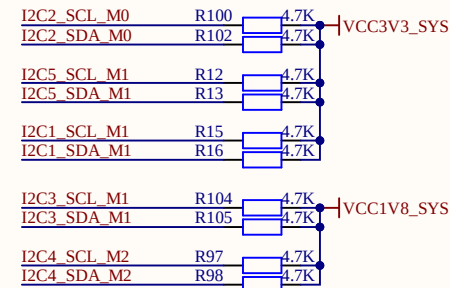
## TF Card



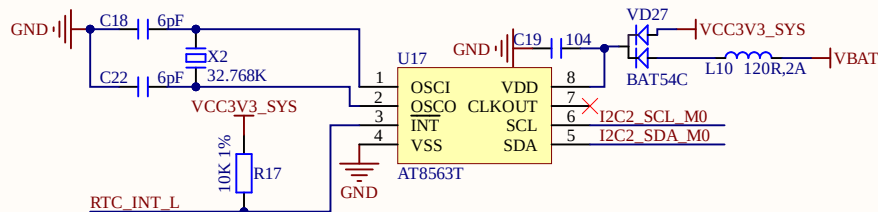
## ADC



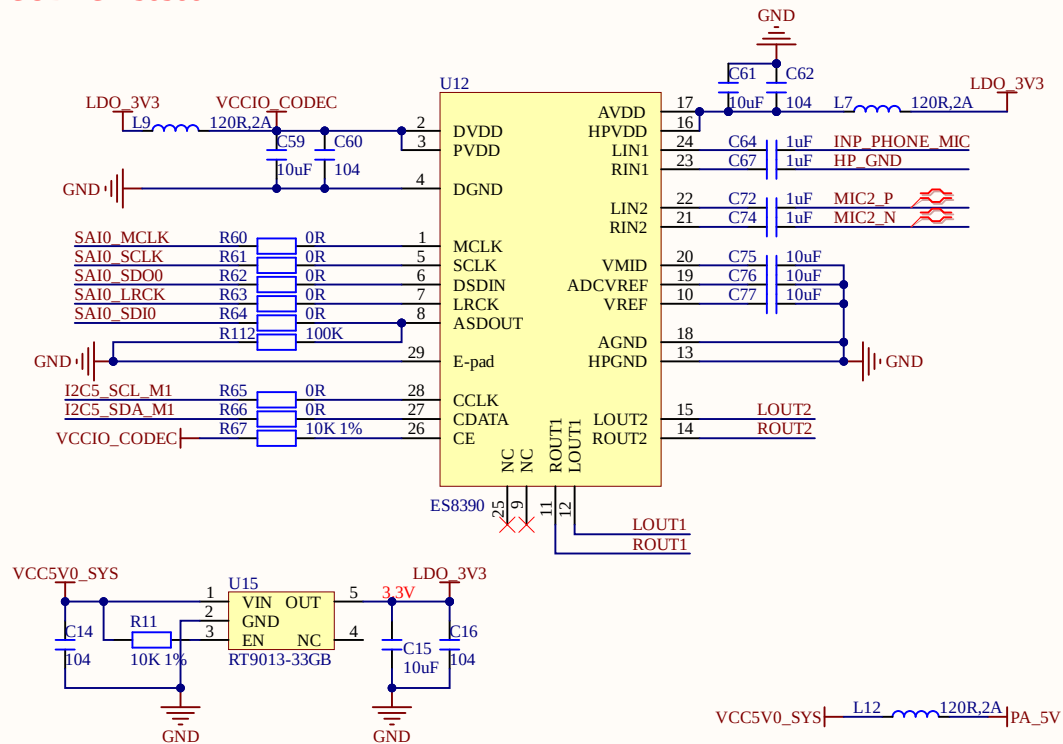
## IIC



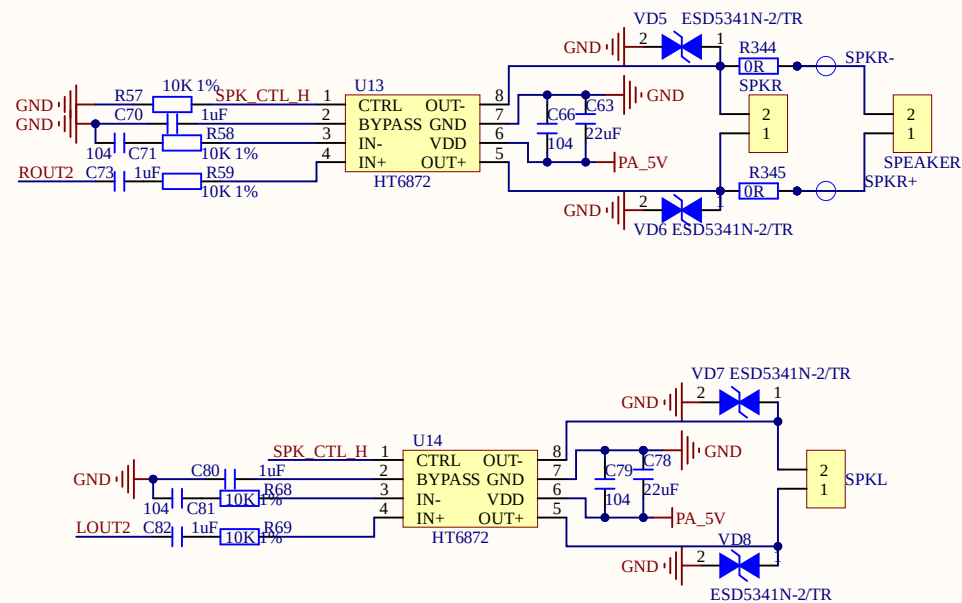
## RTC



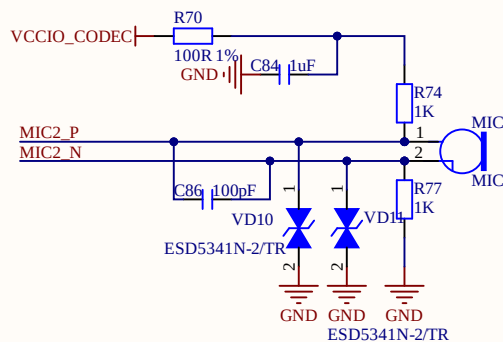
## CODEC ES8388



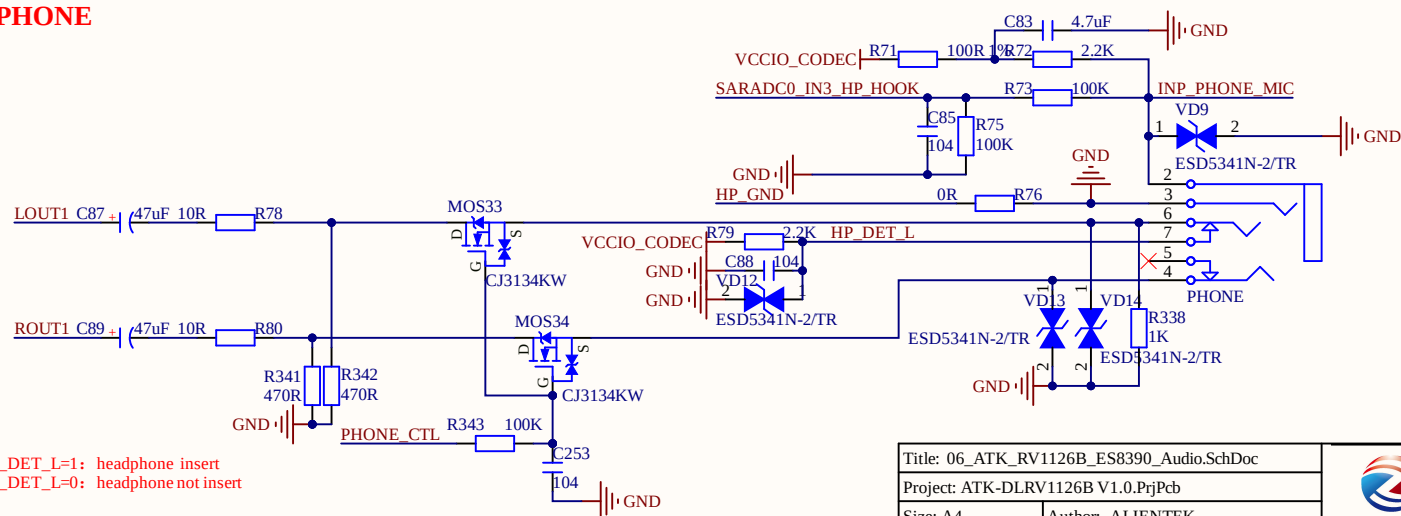
## SPEAKER



## MIC



## EARPHONE



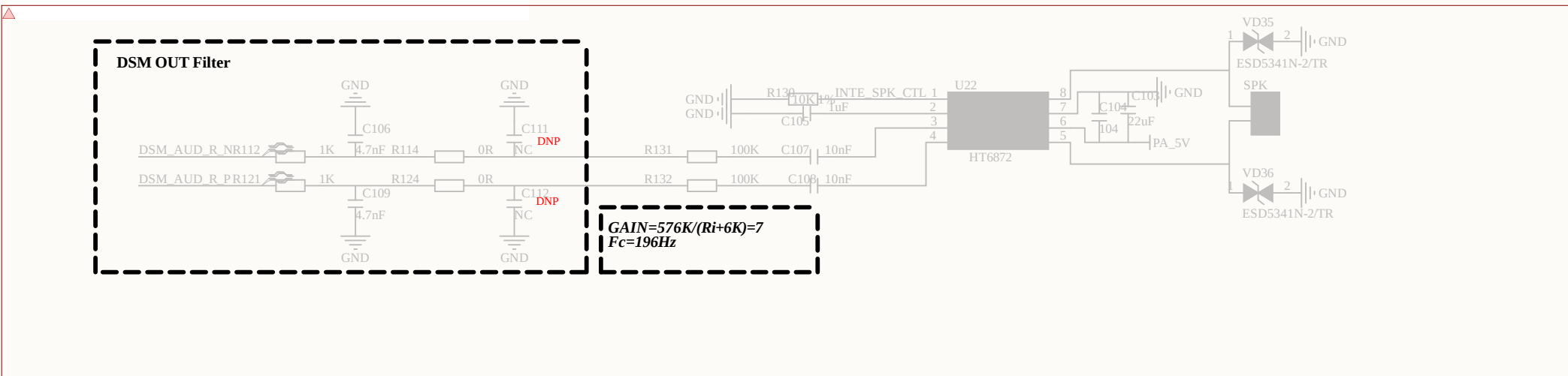
HP\_DET\_L=1: headphone insert  
HP\_DET\_L=0: headphone not insert

|                                           |                  |
|-------------------------------------------|------------------|
| Title: 06_ATK_RV1126B_ES8390_Audio.SchDoc |                  |
| Project: ATK-DLRV1126B V1.0.PriPcb        |                  |
| Size: A4                                  | Author: ALIENTEK |
| Date: 2025-12-31                          | Version: V1.0    |
| Sheet: 6 of 13                            |                  |

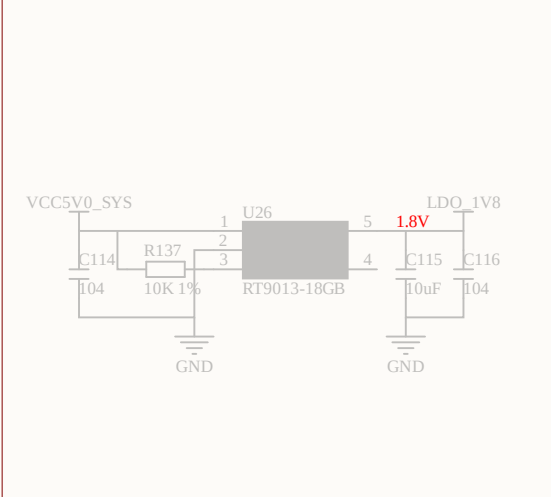


## SPEAKER

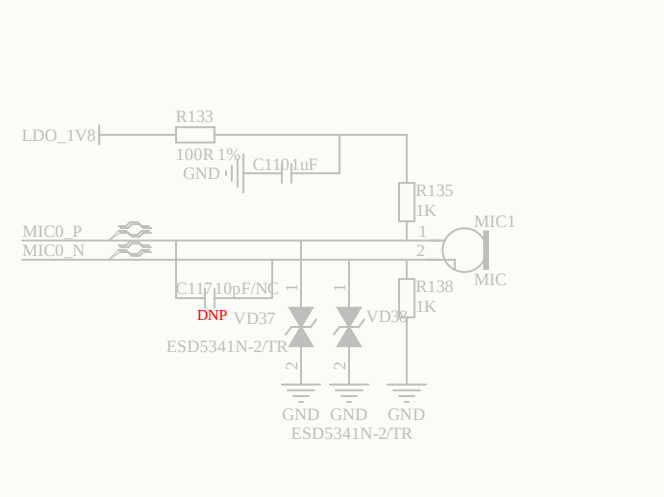
可选设计！RV1126B内部音频接口，只在原理图上做参考设计，正点原子RV1126B开发板实际使用的外置ES8388，并没有使用RV1126B内部音频！



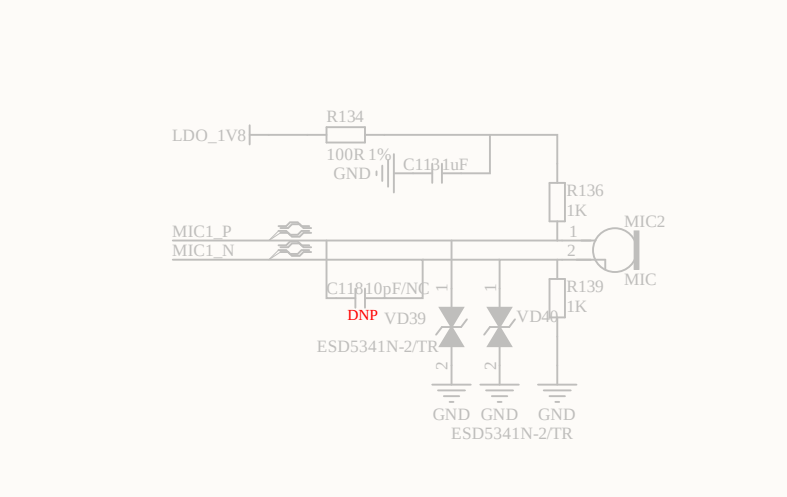
## AUDIO 1.8V



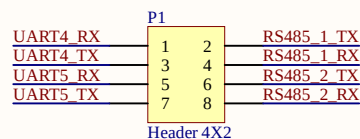
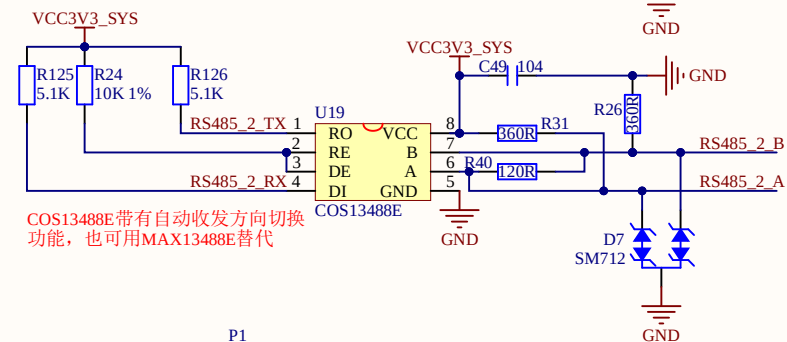
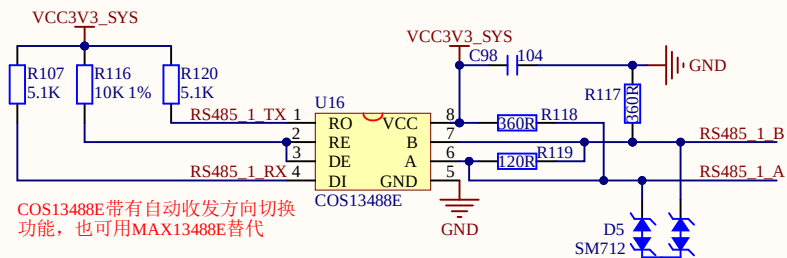
## ADC0-MIC IN



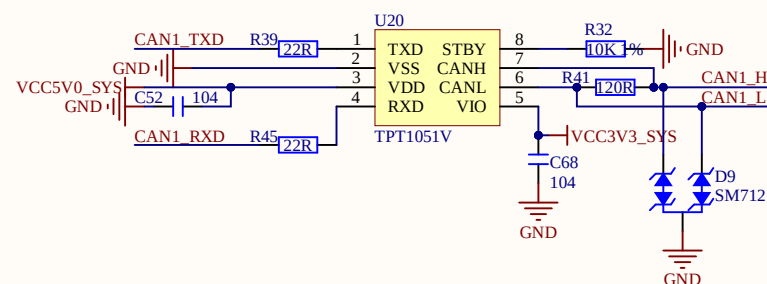
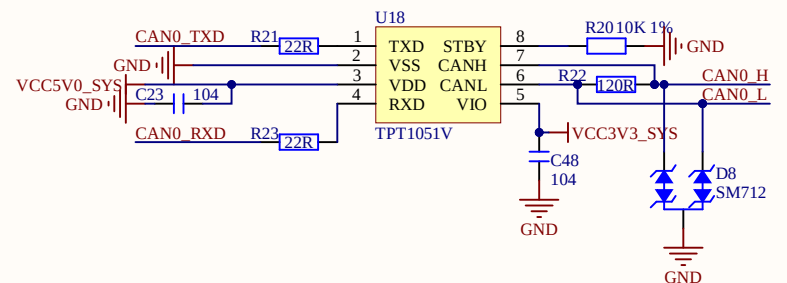
## ADC1-MIC IN



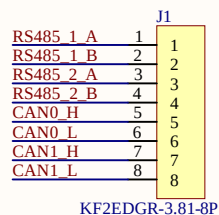
## RS485



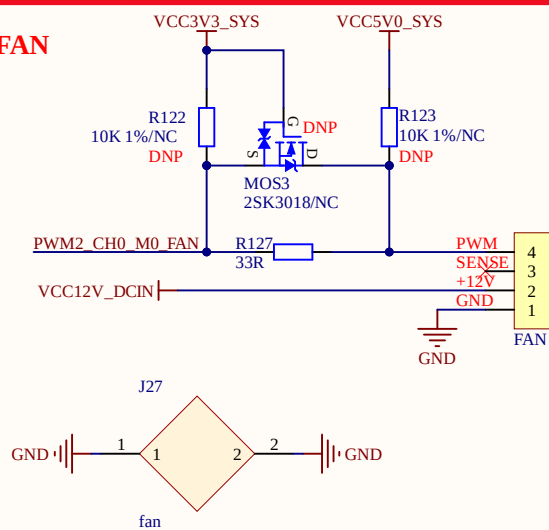
## CAN



## FDCAN&RS485 HEADER



## FAN

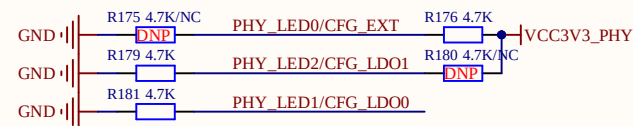
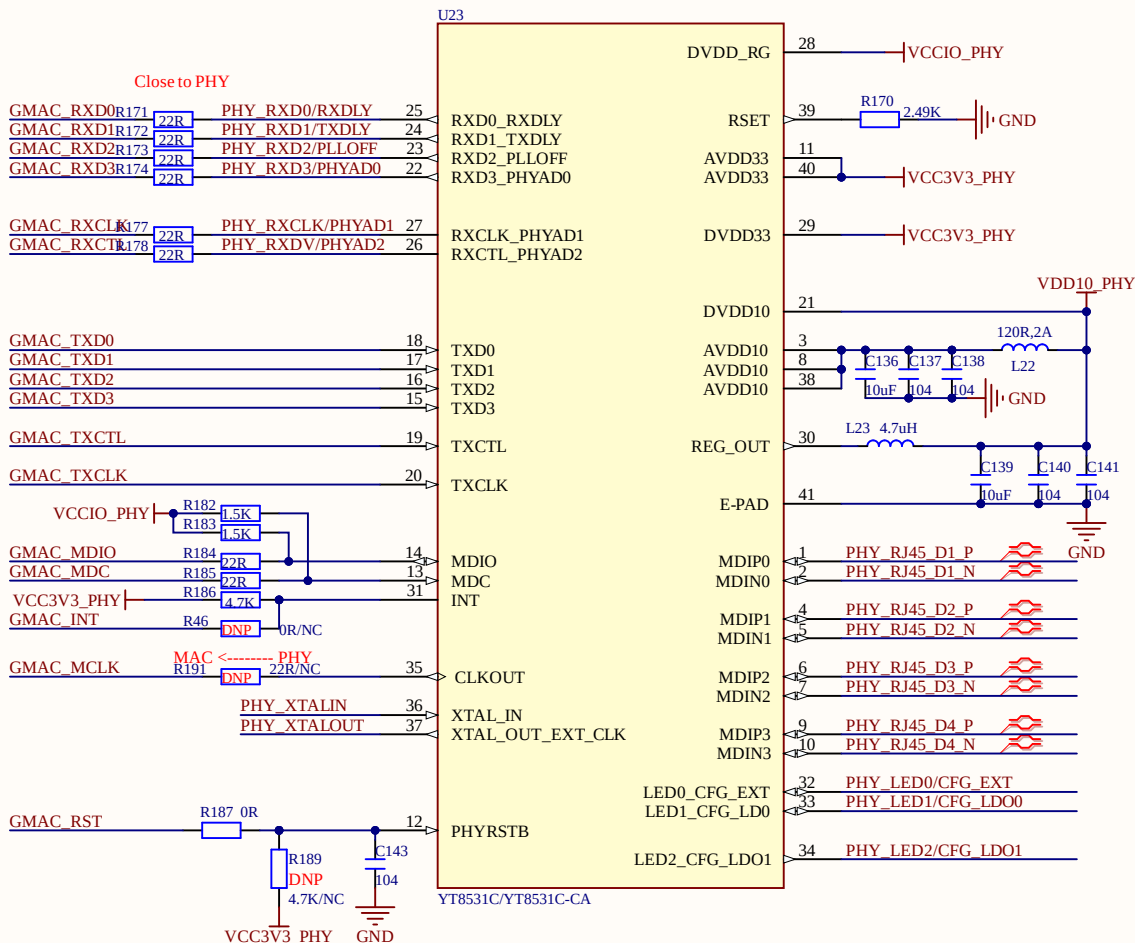
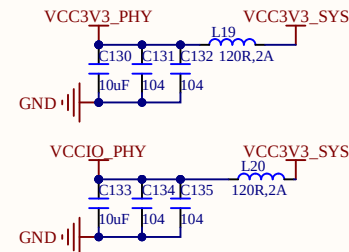
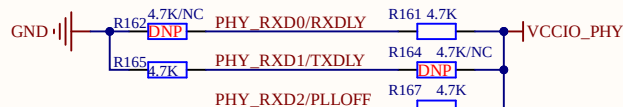
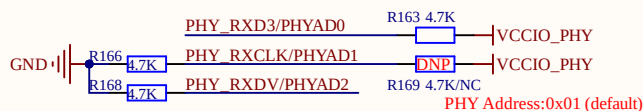


|                                        |                  |                |
|----------------------------------------|------------------|----------------|
| Title: 08_ATK_RV1126B_RS485_CAN.SchDoc |                  |                |
| Project: ATK-DLVRV1126B V1.0.PrjPcb    |                  |                |
| Size: A4                               | Author: ALIENTEK |                |
| Date: 2025-12-31                       | Version: V1.0    | Sheet: 8 of 13 |

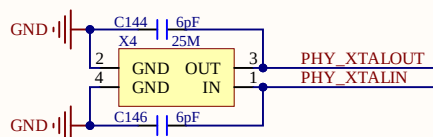
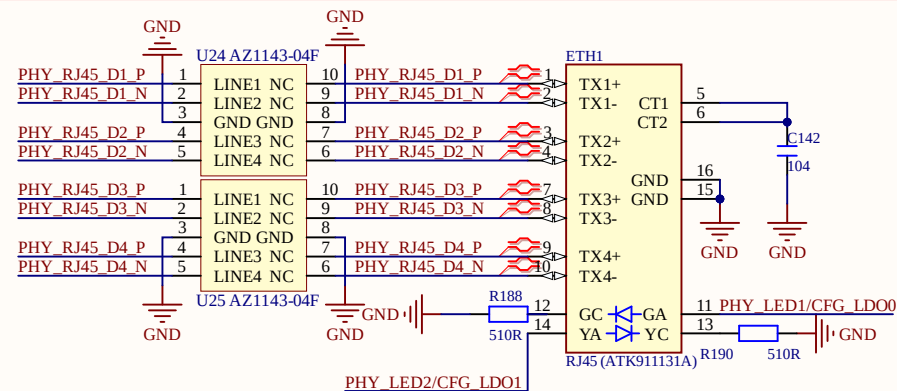




# RGMII 10/100/1000M 3.3V

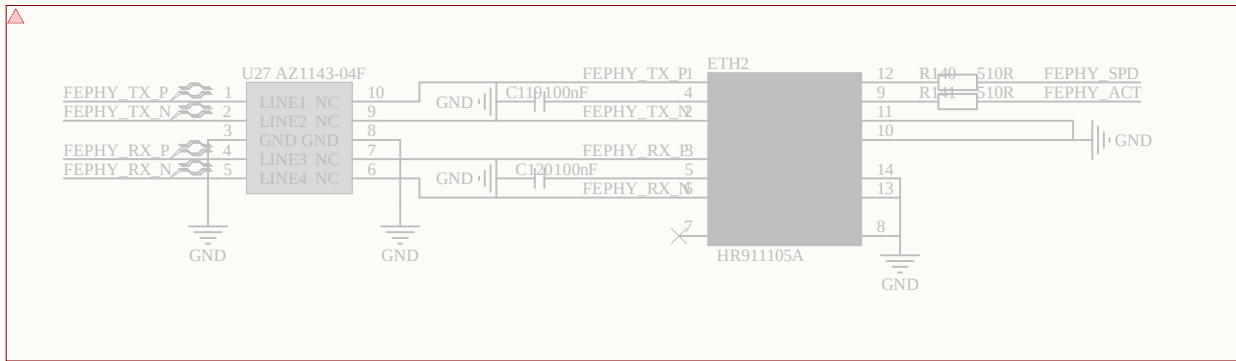


| RGMII Power Source      | CFG_EXT | CFG_LDO[1:0] |
|-------------------------|---------|--------------|
| External 3.3V (default) | 1'b1    | 2'b00        |
| External 1.8V           | 1'b1    | 2'b00        |
| External 1.8V           | 1'b0    | 2'b10        |

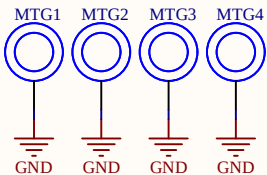


## FEPHY 100M

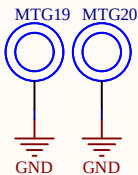
可选设计！RV1126B内部百M网络，只在原理图上做参考设计，正点原子RV1126B开发板实际使用采用外部PHY，实现千M网络，并没有使用RV1126B内部百M网络！



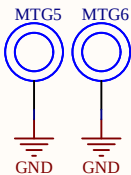
### 核心板固定螺母



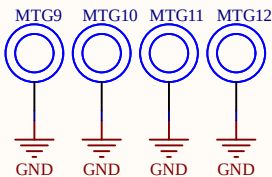
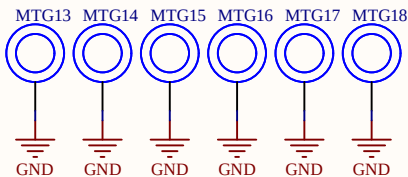
### 4G模块固定螺母



### 液晶模块固定螺母



### 亚克力盖板固定螺母



### MARK点



Title: 10\_ATK\_RV1126B\_FEPHY\_NET\_OPT.SchDoc

Project: ATK-DLRV1126B V1.0.PrjPcb

Size: A4

Author: ALIENTEK

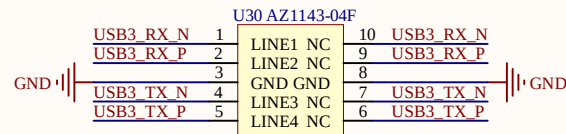
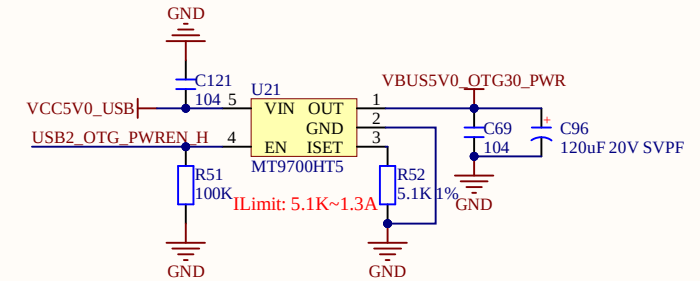
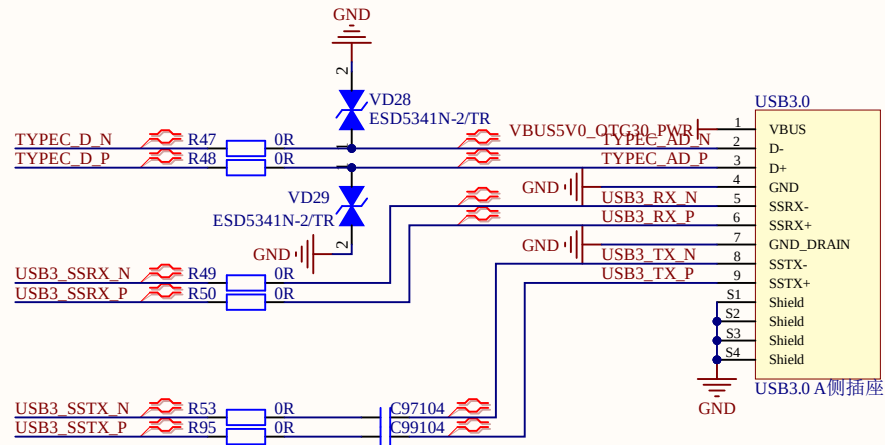
Date: 2025-12-31

Version: V1.0

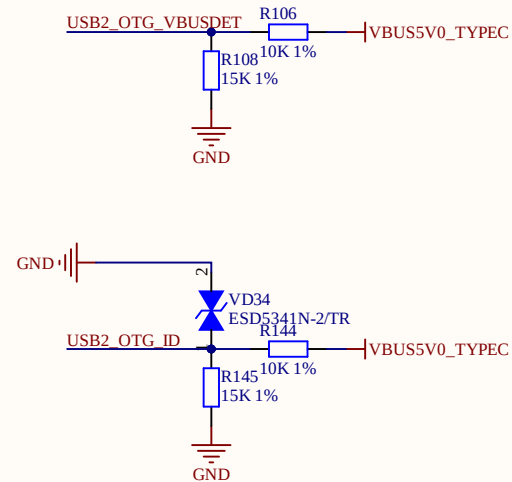
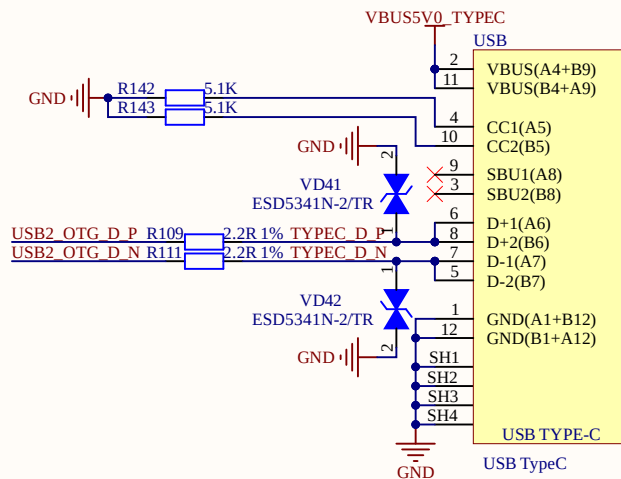
Sheet: 10 of 13



## USB3.0 HOST



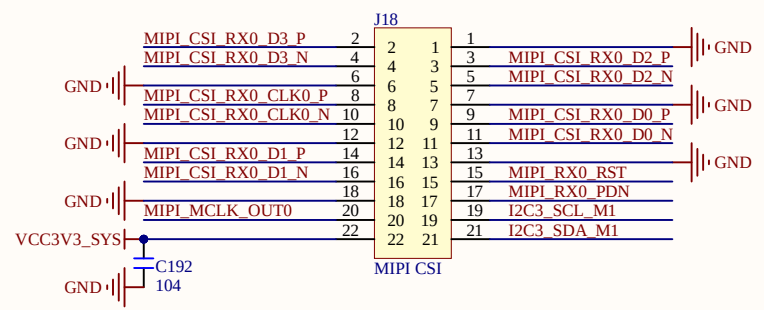
## USB TYPEC



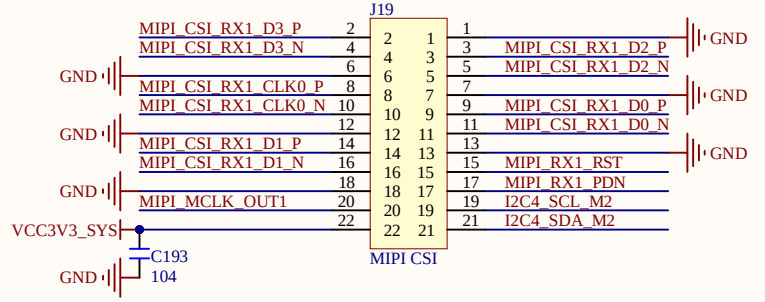
|                                          |                  |                 |
|------------------------------------------|------------------|-----------------|
| Title: 11_ATK_RV1126B_TYPEC_USB30.SchDoc |                  |                 |
| Project: ATK-DLRV1126B V1.0.PrjPcb       |                  |                 |
| Size: A4                                 | Author: ALIENTEK |                 |
| Date: 2025-12-31                         | Version: V1.0    | Sheet: 11 of 13 |



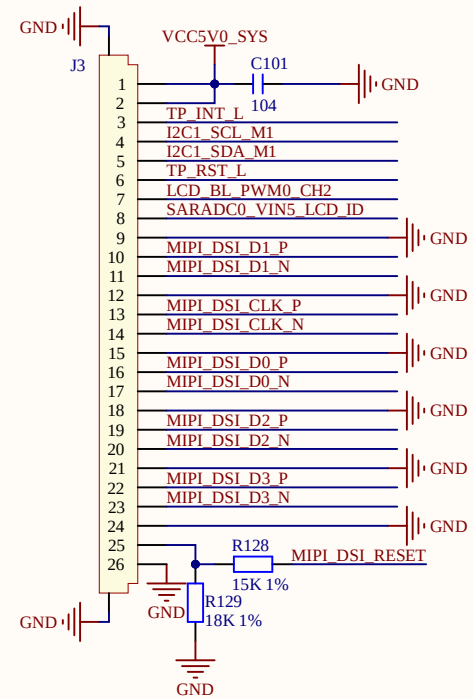
MIPI CSI0



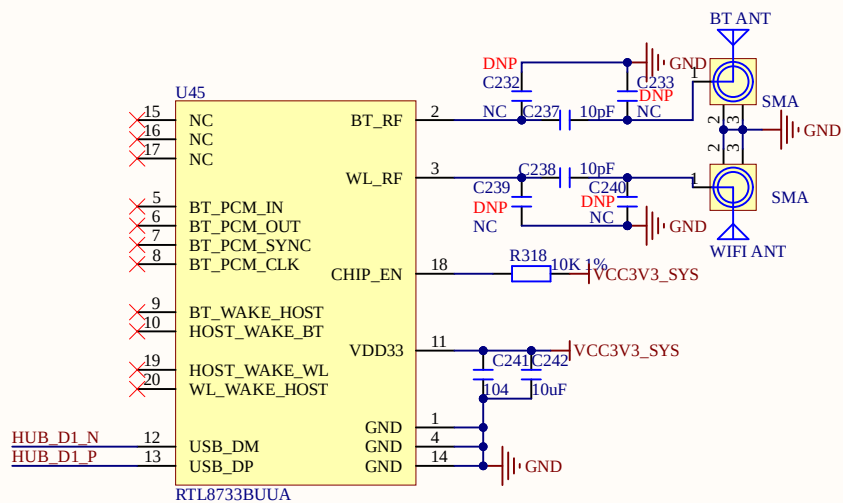
MIPI CSI1



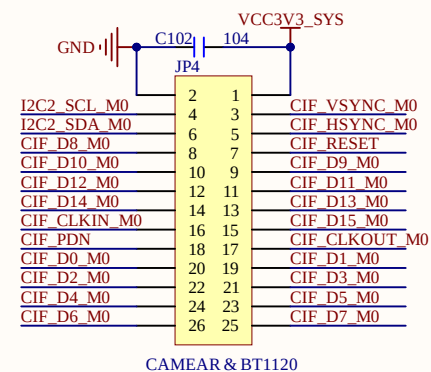
MIPI DSI



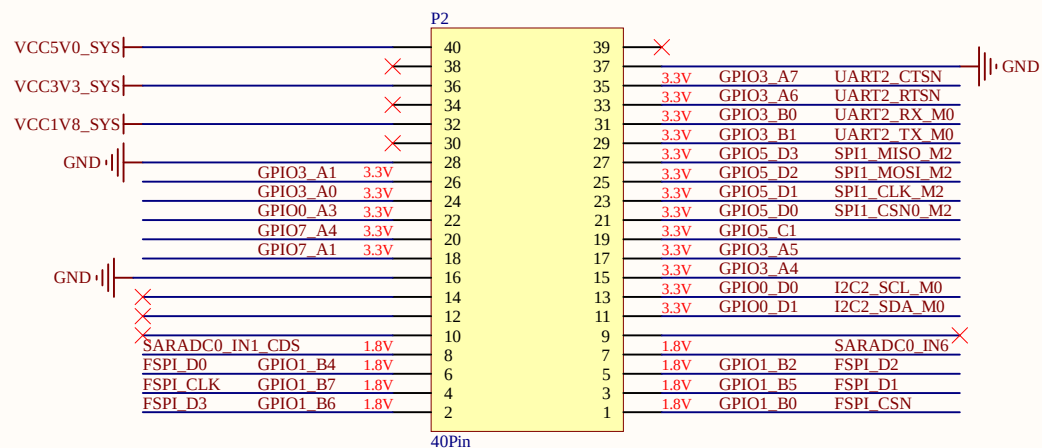
## USB WIFI&BT



## CIF Camera & BT1120



## PINOUT



|                                           |                  |                 |
|-------------------------------------------|------------------|-----------------|
| Title: 13_ATK_RV1126B_WIFI_CIF_PIN.SchDoc |                  |                 |
| Project: ATK-DLRV1126B V1.0.PrjPcb        |                  |                 |
| Size: A4                                  | Author: ALIENTEK |                 |
| Date: 2025-12-31                          | Version: V1.0    | Sheet: 13 of 13 |

